WAIMLAp 2026

Workshop on Artificial Intelligence and Machine Learning Applications

PUCP

Adapting Segment Anything for Sentinel-1 Water Mapping with Automatic Prompts

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MOTIVATION & RESEARCH PROBLEM

- Sentinel-1 radar sees surface water through the clouds of extreme rainfall, as in the 2017 coastal El Niño floods in Piura, Peru.
- State of the art on Sen1Floods11 [1] with radar and hand labels: global IoU 0.68 [2]. Reaching 0.72 takes an optical teacher [2] or auxiliary data and extra weak labels [3].
- SAM [4] was pretrained on natural images, far from SAR [5]. LoRA [6] helps on other SAR data [7], but we found no published SAM result on these chips with radar only.
- **Research question:** can SAM map water from radar with automatic prompts, and what must be adapted?

OBJECTIVES

General objective

To evaluate whether an adapted SAM can map surface water on Sentinel-1 imagery using only automatic prompts.

Specific objectives

- To identify which SAM component limits performance on SAR.
- To adapt input, encoder and decoder to reach a global IoU of 0.60.
- To compare the adapted model with threshold and U-Net baselines on held-out data and an unseen region.

METHODOLOGY

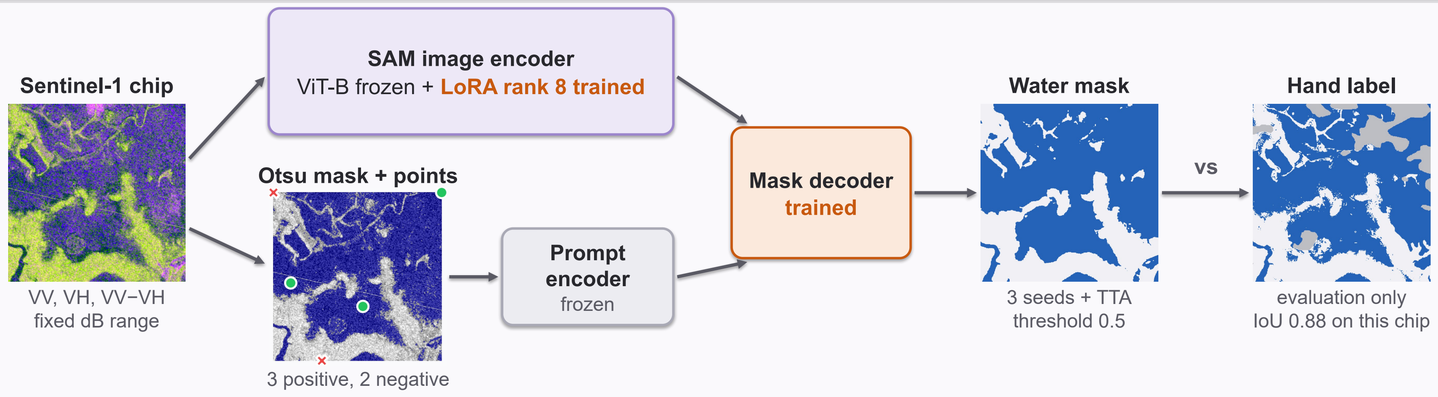


Fig. 1. Final pipeline on a Bolivia chip. Prompts come from Otsu, so inference needs no labels. Only LoRA and the mask decoder are trained.

Table 1. Experimental setup

Data	Sen1Floods11 v1.1 [1], hand labels, 512 × 512 px: 250 train, 86 valid, 89 test, 15 Bolivia
Prompts	Otsu [8] on VH; no labels
Model	SAM ViT-B [4]; LoRA on 12 encoder blocks [9] + mask decoder (4.35 M trainable)
Training	15 epochs, AdamW, lr 1e-4, BCE + Dice, D4 flips and rotations; no hyperparameter search
Baselines	Otsu; fixed VH threshold −23.3 dB + 5×5 median [11]†; U-Net from scratch [10], 7.76 M
Metrics	Global IoU (pooled TP, FP, FN); mean per-chip IoU; paired bootstrap 95 % CI; Wilcoxon
Protocol	Pre-registered steps; test and Bolivia evaluated once

RESULTS & DISCUSSION

R1. Pretrained SAM fails on radar; better prompts do not fix it

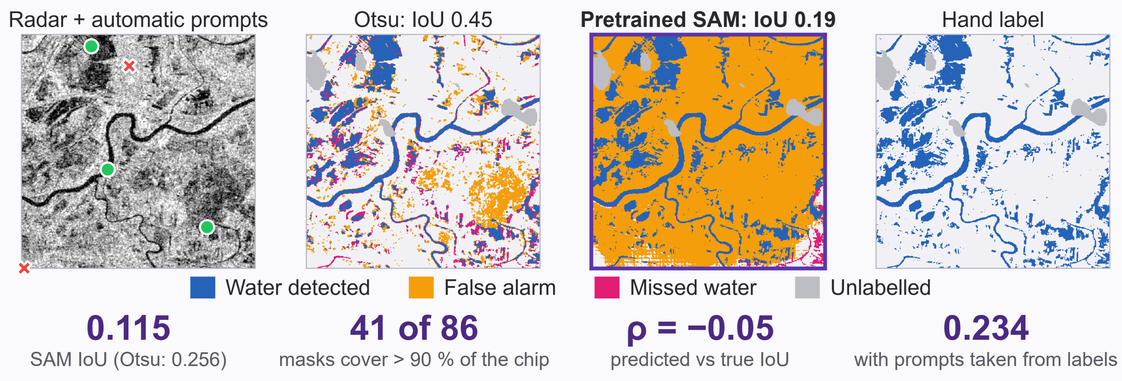


Fig. 2. Validation chip with 19 % water: pretrained SAM marks 95 % as water.

R2. Adapting input, encoder and decoder lifts validation IoU to 0.615

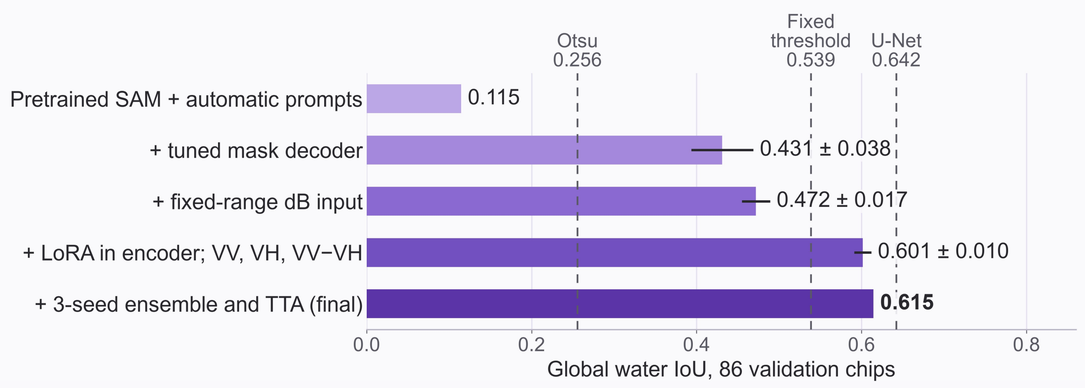


Fig. 3. Each bar adds one change (± SD over 3 seeds). Validation guided these choices, so values are optimistic.

R3. On held-out data, SAM + LoRA exceeds the 0.60 target

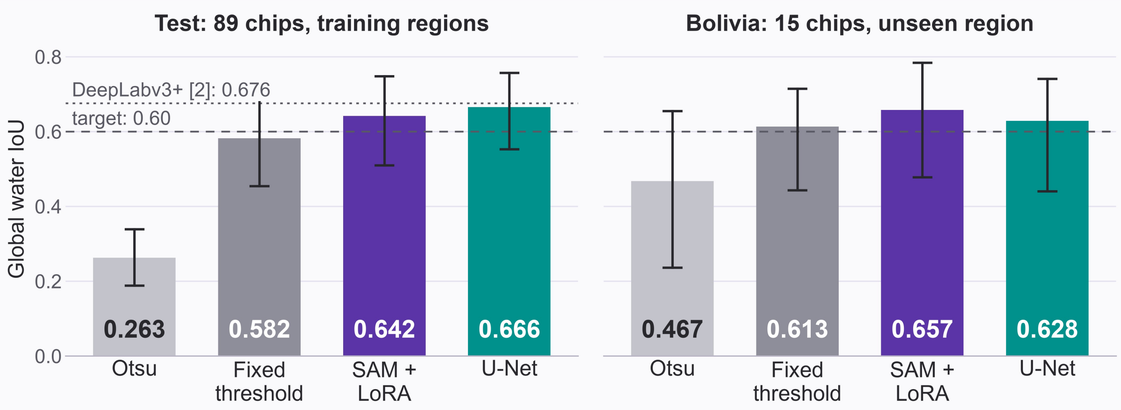


Fig. 4. Global IoU, evaluated once; whiskers show the 95 % chip-bootstrap CI. DeepLabv3+ [2] was measured at 16 m.

R4. U-Net wins on test, SAM on the unseen region, by small margins

Table 2. Final results	Test (89 chips)				Bolivia (15 chips)			
Method	**IoU**	**Mean**	**Prec.**	**Recall**	**IoU**	**Mean**	**Prec.**	**Recall**
Otsu	0.263	0.235	0.300	0.676	0.467	0.427	0.513	0.840
Fixed threshold†	0.582	0.314	0.812	0.673	0.613	0.371	0.914	0.651
SAM + LoRA (ours)	**0.642**	**0.405**	**0.836**	**0.734**	**0.657**	**0.438**	**0.834**	**0.757**
U-Net	0.666	0.421	0.814	0.785	0.628	0.397	0.849	0.708

Table 3. Paired differences in global IoU	Test [95 % CI]	Bolivia [95 % CI]
SAM − threshold	+0.059 [+0.026, +0.101]	+0.044 [−0.005, +0.092]
U-Net − threshold	+0.083 [+0.053, +0.120]	+0.015 [−0.025, +0.057]
U-Net − SAM	+0.024 [−0.007, +0.061]	−0.029 [−0.069, +0.021]

Mean: mean per-chip IoU. It is lower because near-dry chips (< 1 % water) are 31 % of test chips but 3 % of the pixels, with IoU 0.15.

†Chosen on train; added after test was opened. **Wilcoxon, U-Net vs SAM:** test p = 0.011, Bolivia p = 0.046. One test chip has no radar data and stays empty for every method.

R5. Where it works and where it fails

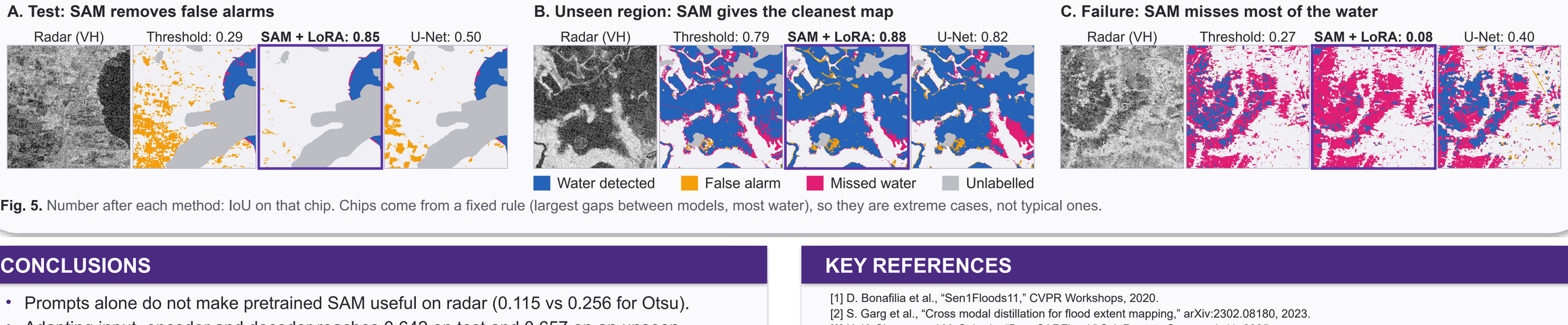


Fig. 5. Number after each method: IoU on that chip. Chips come from a fixed rule (largest gaps between models, most water), so they are extreme cases, not typical ones.

CONCLUSIONS

- Prompts alone do not make pretrained SAM useful on radar (0.115 vs 0.256 for Otsu).
- Adapting input, encoder and decoder reaches 0.642 on test and 0.657 on an unseen region, with radar only and 250 training chips.
- A U-Net from scratch beats it in the training regions and is more stable; on the unseen region, a fixed threshold comes close.
- **Limits:** 15 chips from a single unseen region, wide intervals, no Peru data yet.

Future work: Piura 2017 test against Copernicus EMS maps; optical data during training; SAR-pretrained models.

KEY REFERENCES

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[3] N. K. Sharma and M. Saharia, “DeepSARFlood,” Sci. Remote Sens., vol. 11, 2025.
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[11] sar-flood-unet, github.com/Ronnelpabalan/sar-flood-unet.

ACKNOWLEDGMENTS / FUNDING

INF659 course project, PUCP. Model: Meta SAM ViT-B (Apache-2.0). Data: Sen1Floods11 (Cloud to Street) and Copernicus Sentinel-1. Compute: one desktop GPU (RTX 3070). No external funding. AI coding and writing assistants were used; each use is logged in the project's decision log.

CONTINUE THE CONVERSATION

Code, results and decision log:
github.com/LuisMirandaMallqui/INF659_TareaAcademica_SAM
Contact: luis.mirandam@pucp.edu.pe
Ask us: would you adapt SAM or just train a U-Net?

BRING YOUR RESEARCH. SHARE YOUR IDEAS. JOIN THE CONVERSATION.
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